

DATABASE MANAGEMENT SYSTEM (DBMS) – TAKE HOME

TEST REPORT

ONLINE RETAIL MANAGEMENT SYSTEM

BY

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I designed the Online Retail Management System to help run a basic online shop. It handles customer details product categories, tracking products in stock, managing customer orders, and recording payments.

1. System Description

This system is a digital version of a physical retail store, which makes data much easier to access, cuts down on duplicated records, and makes the shopping process more organized.

2. Tables Created

The database has six tables:

- 1. Customers:** Holds customer info like names, emails, phone numbers, and addresses.
- 2. Categories:** Groups products into types (e.g., Electronics, Clothing).
- 3. Products:** Keeps track of items for sale, their prices, and stock numbers.
- 4. Orders:** Records customer orders, order dates, totals, and if they are delivered, shipped, or pending.
- 5. OrderItems:** Connects orders and products (this handles cases where a customer buys multiple products in one order).
- 6. Payments:** Records payments made for each order.

To make sure all the tables connect properly and there are no orphan records, I set up these relationships using foreign keys:

3. Relationships Used

- **Customers and Orders:** One customer can place many orders. The CustomerID in the Orders table links back to the Customers table.
- **Categories and Products:** One category can contain many products. The CategoryID in the Products table links to the Categories table.
- **Orders and OrderItems:** One order can contain many different items. The OrderID in the OrderItems table links to the Orders table.

- **Products and OrderItems:** A product can show up in many different order items. The ProductID in the OrderItems table links to the Products table.
- **Orders and Payments:** An order can have payments. The OrderID in the Payments table links to the Orders table.

4. Advanced Database Features

View (OrderSummary): I created a view named OrderSummary. This view joins the Customers, Orders, OrderItems, and Products tables so you can easily see order details without writing long JOIN queries every time.

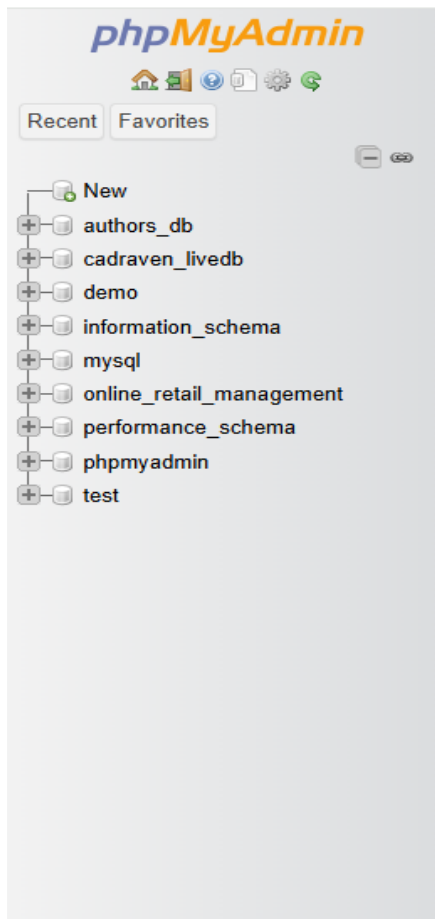
Trigger (After_OrderItem_Insert): I also wrote a trigger called After_OrderItem_Insert. Every time a new product item is sold (inserted into OrderItems), this trigger automatically subtracts the quantity from the stock in the Products table, keeping our inventory accurate.

5. Screenshots of Database Operations

Since I already took the screenshots and saved them in the project folder, here they are showing the database running in phpMyAdmin.

5.1. Database Created

Here is the screenshot showing that the **online_retail_management** database was created successfully.



5.2. Tables created

This shows the tables created in the database.

phpMyAdmin

Server: 127.0.0.1 Database: online_retail_management

Structure SQL Search Query Export Import Operations Privileges Routines Events More

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 - orders
 - payments
 - products
 - Views
- performance_schema
- phpmyadmin
- test

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ categories		5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ customers		20	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ orderitems		21	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ orders		20	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ payments		20	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ products		20	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	106	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

☐ Check all With selected:

Print Data dictionary

Create new table

Table name Number of columns

Console

Activate Windows
Go to Settings to activate Windows.

5.3. Inserted records

This shows some of the customer records that were successfully inserted.

Showing rows 0 - 19 (20 total, Query took 0.0005 seconds)

```
SELECT * FROM `customers`
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	CustomerID	FirstName	LastName	Email	Phone	Address	RegistrationDate
☐	1	Racheal	Nalweyiso	nalweyisoracheal@gmail.com	+256701234567	Ntinda, Kampala, Uganda	2028-05-23 22:38:21
☐	2	Bob	Muwanguzi	bobmuwanguzi@gmail.com	+256702345678	Kololo, Kampala, Uganda	2028-05-23 22:38:21
☐	3	Charlie	Okello	charlieokello@gmail.com	+256703456789	Gulu City, Uganda	2028-05-23 22:38:21
☐	4	David	Williams	davidwilliams@gmail.com	+256704567890	Entebbe, Wakiso, Uganda	2028-05-23 22:38:21
☐	5	Eva	Abagye	evaabagye@gmail.com	+256705678901	Mbarara City, Uganda	2028-05-23 22:38:21
☐	6	Frank	Kagalama	frankkagalama@gmail.com	+254701234567	Westlands, Nairobi, Kenya	2028-05-23 22:38:21
☐	7	Grace	Akello	graceakello@gmail.com	+254702345678	Kisumu, Kenya	2028-05-23 22:38:21
☐	8	Henry	Davis	henrydavis@gmail.com	+254703456789	Karen, Nairobi, Kenya	2028-05-23 22:38:21
☐	9	Ivy	Namakula	ivynamakula@gmail.com	+254704567890	Nakuru, Kenya	2028-05-23 22:38:21
☐	10	Jack	Ssemanda	jackssemanda@gmail.com	+254705678901	Mombasa, Kenya	2028-05-23 22:38:21
☐	11	Karen	Nankya	karennankya@gmail.com	+250781234567	Kigali City, Rwanda	2028-05-23 22:38:21
☐	12	Leo	Kintu	leokintu@gmail.com	+250782345678	Gasabo, Kigali, Rwanda	2028-05-23 22:38:21
☐	13	Mia	Kyomuwendo	miakyomuwendo@gmail.com	+250783456789	Huye, Rwanda	2028-05-23 22:38:21
☐	14	Noah	Mbaguta	noahmbaguta@gmail.com	+250784567890	Musanze, Rwanda	2028-05-23 22:38:21
☐	15	Olivia	Namata	olivianamata@gmail.com	+250785678901	Rubavu, Rwanda	2028-05-23 22:38:21
☐	16	Paul	Thomas	paulthomas@gmail.com	+447911123456	Manchester, United Kingdom	2028-05-23 22:38:21
☐	17	Quinn	Ntale	quinnntale@gmail.com	+447922234567	Birmingham, United Kingdom	2028-05-23 22:38:21
☐	18	Rachel	Mutoni	rachelmutoni@gmail.com	+447933345678	London, United Kingdom	2028-05-23 22:38:21
☐	19	Sam	Bagenda	sambagenda@gmail.com	+447944456789	Liverpool, United Kingdom	2028-05-23 22:38:21
☐	20	Tina	Mbabazi	tinababazi@gmail.com	+447955567890	Leeds, United Kingdom	2028-05-23 22:38:21

5.4. Query outputs

Here are the screenshots showing my queries executing successfully.

Query 1: Running basic queries and showing table outputs.

Showing rows 0 - 19 (20 total, Query took 0.0009 seconds)

```
SELECT * FROM customers;
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	CustomerID	FirstName	LastName	Email	Phone	Address	RegistrationDate
☐	1	Racheal	Nalweyiso	nalweyisoracheal@gmail.com	+256701234567	Ntinda, Kampala, Uganda	2026-05-23 22:36:21
☐	2	Bob	Muwanguzi	bobmuwanguzi@gmail.com	+256702345678	Kololo, Kampala, Uganda	2026-05-23 22:36:21
☐	3	Charlie	Okello	charlieokello@gmail.com	+256703456789	Gulu City, Uganda	2026-05-23 22:36:21
☐	4	David	Williams	davidwilliams@gmail.com	+256704567890	Entebbe, Wakiso, Uganda	2026-05-23 22:36:21
☐	5	Eva	Abagye	evaabagye@gmail.com	+256705678901	Mbarara City, Uganda	2026-05-23 22:36:21
☐	6	Frank	Kagalama	frankkagalama@gmail.com	+254701234567	Westlands, Nairobi, Kenya	2026-05-23 22:36:21
☐	7	Grace	Akello	graceakello@gmail.com	+254702345678	Kisumu, Kenya	2026-05-23 22:36:21
☐	8	Henry	Davis	henrydavis@gmail.com	+254703456789	Karen, Nairobi, Kenya	2026-05-23 22:36:21
☐	9	Ivy	Namakula	ivynamakula@gmail.com	+254704567890	Nakuru, Kenya	2026-05-23 22:36:21
☐	10	Jack	Ssemanda	jackssemanda@gmail.com	+254705678901	Mombasa, Kenya	2026-05-23 22:36:21
☐	11	Karen	Nankya	karennankya@gmail.com	+250781234567	Kigali City, Rwanda	2026-05-23 22:36:21
☐	12	Leo	Kintu	leokintu@gmail.com	+250782345678	Gasabo, Kigali, Rwanda	2026-05-23 22:36:21

Query 2: Retrieving products above a certain price.

The screenshot shows the phpMyAdmin interface with the 'categories' table selected. The table structure is displayed, showing columns: CategoryID, CategoryName, and Description. The table contains 5 rows of data.

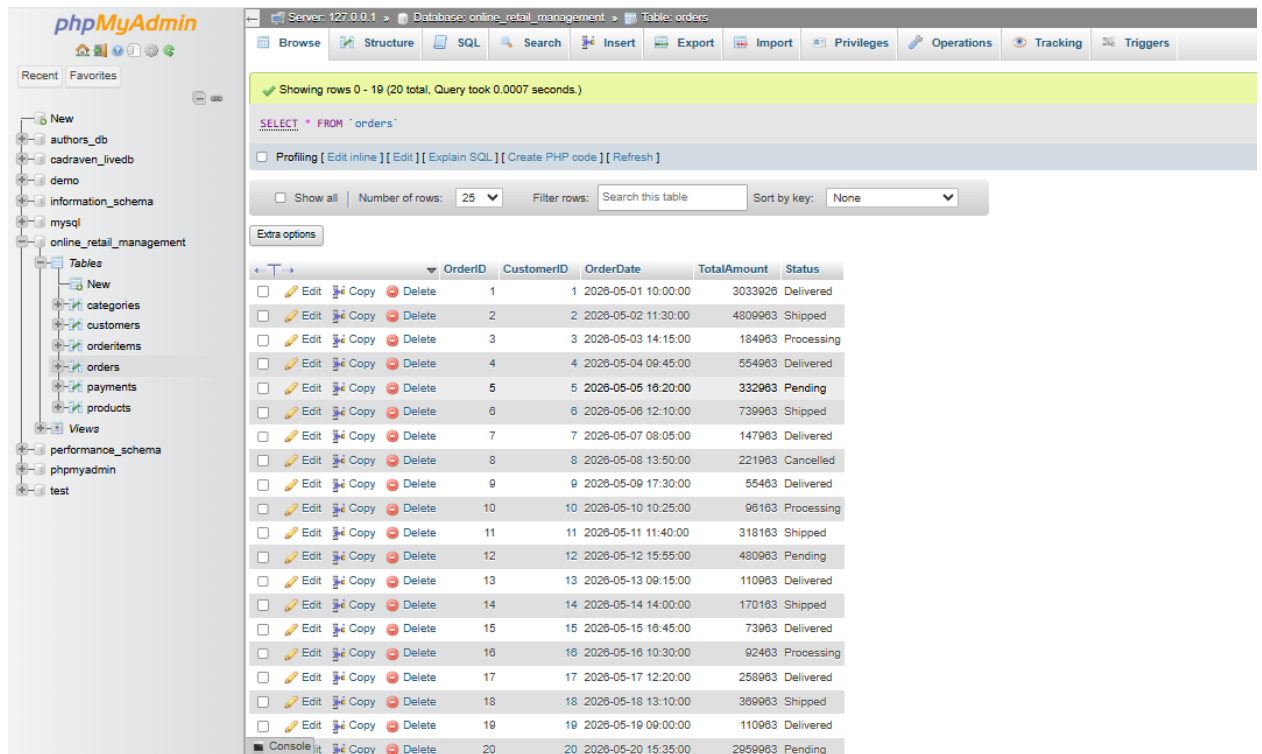
CategoryID	CategoryName	Description
1	Electronics	Devices and gadgets
2	Clothing	Men and Women fashion
3	Home & Kitchen	Home appliances and furniture
4	Books	Physical and E-books
5	Sports	Sporting goods and equipment

Query 3: Ordering data.

The screenshot shows the phpMyAdmin interface with the 'orderitems' table selected. The table structure is displayed, showing columns: OrderItemID, OrderID, ProductID, Quantity, and UnitPrice. The table contains 21 rows of data.

OrderItemID	OrderID	ProductID	Quantity	UnitPrice
1	1	1	1	2959993
2	1	5	1	73993
3	2	2	1	4809993
4	3	6	1	184993
5	4	3	1	554993
6	5	7	1	332993
7	6	4	1	739993
8	7	9	1	147993
9	8	10	1	221993
10	9	13	1	55493
11	10	17	1	96163
12	11	19	1	318163
13	12	12	1	480963
14	13	14	1	110993
15	14	18	1	170163
16	15	16	1	73993
17	16	15	1	92463
18	17	8	1	258993
19	18	11	1	369993
20	19	20	1	110963

Query 4: Running aggregate functions (Sums and Averages).



Server: 127.0.0.1 Database: online_retail_management Table: orders

Showing rows 0 - 19 (20 total, Query took 0.0007 seconds.)

`SELECT * FROM `orders``

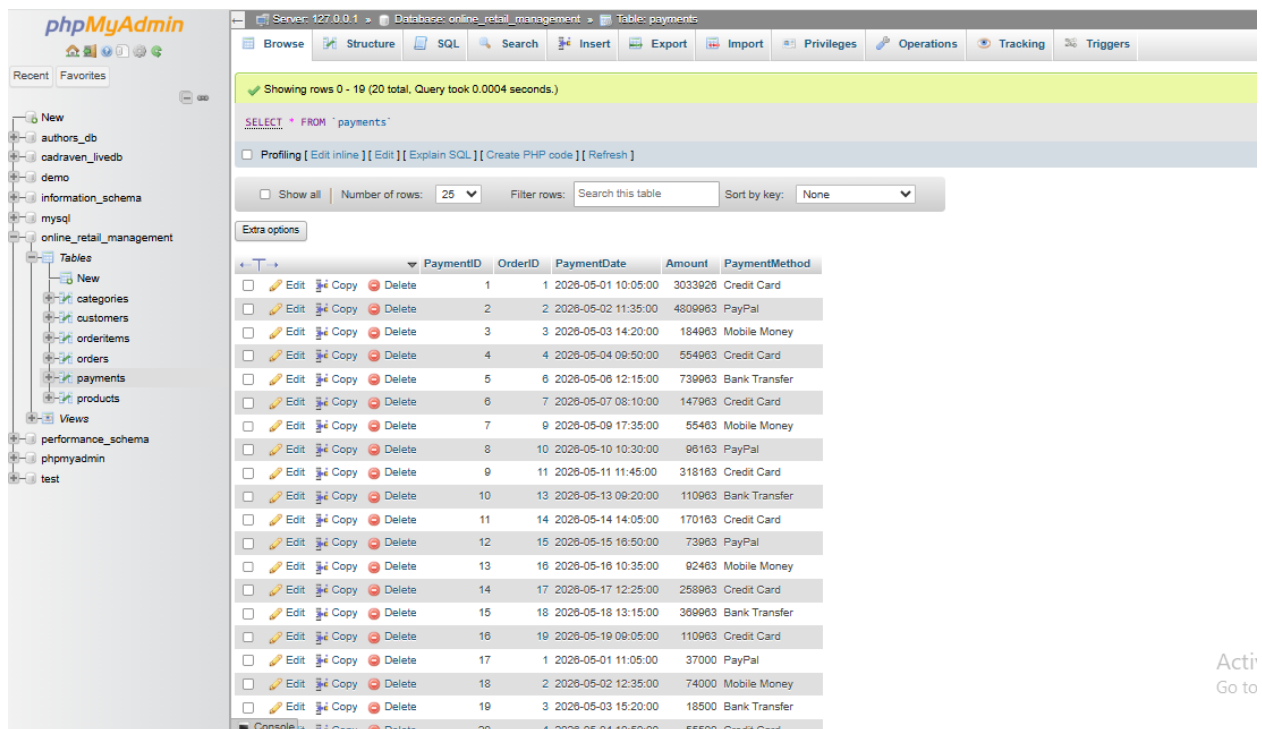
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	OrderID	CustomerID	OrderDate	TotalAmount	Status
☐ Edit ☐ Copy ☐ Delete	1	1	2020-05-01 10:00:00	3033926	Delivered
☐ Edit ☐ Copy ☐ Delete	2	2	2020-05-02 11:30:00	4809963	Shipped
☐ Edit ☐ Copy ☐ Delete	3	3	2020-05-03 14:15:00	184963	Processing
☐ Edit ☐ Copy ☐ Delete	4	4	2020-05-04 09:45:00	554963	Delivered
☐ Edit ☐ Copy ☐ Delete	5	5	2020-05-05 18:20:00	332963	Pending
☐ Edit ☐ Copy ☐ Delete	6	6	2020-05-06 12:10:00	739963	Shipped
☐ Edit ☐ Copy ☐ Delete	7	7	2020-05-07 08:05:00	147963	Delivered
☐ Edit ☐ Copy ☐ Delete	8	8	2020-05-08 13:50:00	221963	Cancelled
☐ Edit ☐ Copy ☐ Delete	9	9	2020-05-09 17:30:00	55463	Delivered
☐ Edit ☐ Copy ☐ Delete	10	10	2020-05-10 10:25:00	96163	Processing
☐ Edit ☐ Copy ☐ Delete	11	11	2020-05-11 11:40:00	318163	Shipped
☐ Edit ☐ Copy ☐ Delete	12	12	2020-05-12 15:55:00	480963	Pending
☐ Edit ☐ Copy ☐ Delete	13	13	2020-05-13 09:15:00	110963	Delivered
☐ Edit ☐ Copy ☐ Delete	14	14	2020-05-14 14:00:00	170163	Shipped
☐ Edit ☐ Copy ☐ Delete	15	15	2020-05-15 18:45:00	73963	Delivered
☐ Edit ☐ Copy ☐ Delete	16	16	2020-05-16 10:30:00	92463	Processing
☐ Edit ☐ Copy ☐ Delete	17	17	2020-05-17 12:20:00	258963	Delivered
☐ Edit ☐ Copy ☐ Delete	18	18	2020-05-18 13:10:00	369963	Shipped
☐ Edit ☐ Copy ☐ Delete	19	19	2020-05-19 09:00:00	110963	Delivered
☐ Edit ☐ Copy ☐ Delete	20	20	2020-05-20 15:35:00	2959963	Pending

Query 5: Running inner joins.



Server: 127.0.0.1 Database: online_retail_management Table: payments

Showing rows 0 - 19 (20 total, Query took 0.0004 seconds.)

`SELECT * FROM `payments``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	PaymentID	OrderID	PaymentDate	Amount	PaymentMethod
☐ Edit ☐ Copy ☐ Delete	1	1	2020-05-01 10:05:00	3033926	Credit Card
☐ Edit ☐ Copy ☐ Delete	2	2	2020-05-02 11:35:00	4809963	PayPal
☐ Edit ☐ Copy ☐ Delete	3	3	2020-05-03 14:20:00	184963	Mobile Money
☐ Edit ☐ Copy ☐ Delete	4	4	2020-05-04 09:50:00	554963	Credit Card
☐ Edit ☐ Copy ☐ Delete	5	6	2020-05-06 12:15:00	739963	Bank Transfer
☐ Edit ☐ Copy ☐ Delete	6	7	2020-05-07 08:10:00	147963	Credit Card
☐ Edit ☐ Copy ☐ Delete	7	9	2020-05-09 17:35:00	55463	Mobile Money
☐ Edit ☐ Copy ☐ Delete	8	10	2020-05-10 10:30:00	96163	PayPal
☐ Edit ☐ Copy ☐ Delete	9	11	2020-05-11 11:45:00	318163	Credit Card
☐ Edit ☐ Copy ☐ Delete	10	13	2020-05-13 09:20:00	110963	Bank Transfer
☐ Edit ☐ Copy ☐ Delete	11	14	2020-05-14 14:05:00	170163	Credit Card
☐ Edit ☐ Copy ☐ Delete	12	15	2020-05-15 18:50:00	73963	PayPal
☐ Edit ☐ Copy ☐ Delete	13	16	2020-05-16 10:35:00	92463	Mobile Money
☐ Edit ☐ Copy ☐ Delete	14	17	2020-05-17 12:25:00	258963	Credit Card
☐ Edit ☐ Copy ☐ Delete	15	18	2020-05-18 13:15:00	369963	Bank Transfer
☐ Edit ☐ Copy ☐ Delete	16	19	2020-05-19 09:05:00	110963	Credit Card
☐ Edit ☐ Copy ☐ Delete	17	1	2020-05-01 11:05:00	37000	PayPal
☐ Edit ☐ Copy ☐ Delete	18	2	2020-05-02 12:35:00	74000	Mobile Money
☐ Edit ☐ Copy ☐ Delete	19	3	2020-05-03 15:20:00	18500	Bank Transfer
☐ Edit ☐ Copy ☐ Delete	20	4	2020-05-04 10:50:00	55500	Credit Card

Query 6: Running the update queries.

Server: 127.0.0.1 > Database: online_retail_management > Table: products

Showing rows 0 - 19 (20 total. Query took 0.0011 seconds.)

`SELECT * FROM `products``

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	ProductID	CategoryID	ProductName	Description	Price	StockQuantity
☐ Edit Copy Delete	1	1	Smartphone X	Latest smartphone model	2959983	50
☐ Edit Copy Delete	2	1	Laptop Pro	High performance laptop	4809983	30
☐ Edit Copy Delete	3	1	Wireless Earbuds	Noise-cancelling earbuds	554983	100
☐ Edit Copy Delete	4	1	Smartwatch	Fitness tracking watch	739983	75
☐ Edit Copy Delete	5	2	Men T-Shirt	Cotton casual t-shirt	66587	200
☐ Edit Copy Delete	6	2	Women Jeans	Slim fit blue jeans	166467	150
☐ Edit Copy Delete	7	2	Winter Jacket	Warm winter coat	299687	60
☐ Edit Copy Delete	8	2	Sneakers	Running shoes	233067	120
☐ Edit Copy Delete	9	3	Coffee Maker	Programmable coffee machine	147983	80
☐ Edit Copy Delete	10	3	Blender	High-speed blender	221983	45
☐ Edit Copy Delete	11	3	Microwave Oven	Compact microwave	369983	25
☐ Edit Copy Delete	12	3	Vacuum Cleaner	Bagless upright vacuum	480983	40
☐ Edit Copy Delete	13	4	Fiction Novel	Bestselling fiction book	55483	300
☐ Edit Copy Delete	14	4	Programming Guide	Learn Python programming	110983	150
☐ Edit Copy Delete	15	4	Cookbook	Healthy recipes	92483	100
☐ Edit Copy Delete	16	4	History Book	World history overview	73983	80
☐ Edit Copy Delete	17	5	Yoga Mat	Non-slip exercise mat	98183	150
☐ Edit Copy Delete	18	5	Dumbbells Set	Adjustable weights	170183	90
☐ Edit Copy Delete	19	5	Tennis Racket	Professional racket	318183	40
☐ Edit Copy Delete	20	5	Soccer Ball	Size 5 match ball	110983	110